Aditya Singh

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Research Profile

My research at CDFD focused on uncovering mechanisms of metabolic reprogramming in Candida glabrata, with particular emphasis on glycosylphosphatidylinositol (GPI)-anchored, membrane-bound aspartyl proteases (yapsins). The goal was to understand how this pathogen survives and evades host defenses. I characterized the interaction of CgYps1 with one of three metabolic proteins involved in the pentose phosphate pathway and performed site-directed mutagenesis and cleavage assays on another interactor involved in glycolysis. These preliminary findings suggest a role for CgYps1 in regulating carbon metabolic flux in *Candida glabrata* under stress conditions.

Research Experience

Centre for DNA Fingerprinting and Diagnostics (CDFD) Hyderabad, India

Project Associate March 2024 – February 2026

- Investigated metabolic reprogramming in C. glabrata under host-mimicking stress conditions by integrating metabolomics with targeted biochemical assays, focusing on glycolysis and pentose phosphate pathway interactions.
- Purified two metabolic interactors of CgYps1 via Ni-NTA affinity chromatography, achieving >90% purity confirmed by SDS-PAGE, for downstream protein interaction and cleavage assays.
- Expressed and purified catalytically active CgYps1 using a *Pichia pastoris* heterologous expression system, employing ammonium sulphate precipitation followed by FPLC/SEC for isolation of the active protease fraction.
- Generated site-directed mutagenesis (SDM) constructs of a glycolytic CgYps1 interactor and performed in vitro binding and cleavage assays to map functional interaction domains.
- Generated dual-species RNA-seq datasets from macrophage-Candida infection models across multiple timepoints, subsequently contributing to downstream pathway enrichment and differential expression analysis.
- Executed in-vivo infection studies using murine models, supporting experimental design, execution, and data collection for virulence assessment.

Independent Projects

scRNA-seq Preprocessing & Analysis Pipeline

Python · Scanpy · AnnData · FastAPI · GitHub Repo

Designed and built a modular, end-to-end single-cell RNA-seq analysis service: QC filtering, normalisation, HVG selection, PCA/UMAP/Leiden clustering, CellTypist annotation, and per-cluster pathway enrichment (gseapy/Enrichr). Architected an AnnData-based data contract preserving raw counts and KNN graph connectivity for downstream compatibility (scVelo, pySCENIC). Built an asynchronous REST API (FastAPI) with background job execution and status polling. Validated on 10x PBMC3k dataset; 15/15 unit tests passing.
Developed with AI coding assistance (Anthropic Claude); all biological decisions, pipeline architecture, parameter choices, and troubleshooting performed independently.

Publication

Role of glycosylphosphatidylinositol (GPI) - anchored aspartyl proteases in the pathobiology of *Candida glabrata*.
Sarowgi, A., **Singh, A.**, & Kaur, R.
Manuscript in preparation

Academic Projects and Internships

Department of Biotechnology, Kakatiya University October 2022 – April 2023
M.Sc. Final-Year Dissertation Project (Advisor: Dr. Radhika Tippani)

Designed a CRISPR/Cas9 knockout vector targeting the *PDS* (Phytoene desaturase) gene in brinjal (*Solanum melongena* L.). Validated the CRISPR/Cas9 system's efficacy for gene editing in *Solanum melongena L.* using *Agrobacterium*-mediated transformation.

Center for Human Security Studies June 2021 – July 2021
Research and Methodology Intern
Conducted a policy-oriented study on health security challenges in India, focusing on systemic gaps in healthcare access, infrastructure, and governance. Gained exposure to research methodology and structured report writing.

Education

National Institute of Technology, Bhopal (NIT-B) Bhopal, India
Master of Technology (M.Tech.) in Biotechnology August 2023 – March 2024
Grade: 72% (~3.2/4.0 GPA) (*Coursework completed; transferred to research position at CDFD*)

Kakatiya University Warangal, India
Master of Science (M.Sc.) in Biotechnology 2021 - 2023
Grade: 72.5% (~3.3/4.0 GPA)

Technical Skills

➢ **Laboratory Skills** [*Cellular Models → Genetic Manipulation → Downstream Analysis*]
Microbial & Plant Tissue Culture, Fungal culture and Mammalian cell line handling (A498 and THP1); THP1-macrophage cell line maintenance (handling and infection of 1.8×10^{10} cells).

Molecular cloning and knockout generation (CRISPR & HR. mediated); Gene-gun & Agrobacterium based plant cell transformation, Electroporation & Calcium chloride based E.coli. Transformation, Lithium acetate based yeast transformation. PCR.

Immunofluorescence imaging and familiarity with fluorescence microscopy workflows (**ZEISS LSM 900**), ELISA, Co-Immuno Precipitation assays, (IMAC) Affinity chromatography & (SEC) FPLC based Protein Purification (**ÄKTA pure™ chromatography system**), Native and SDS PAGE, Western Blotting, FACS.

➢ **Computational Skills**
LC-MS/MS data analysis, AlphaFold, PyMOL, AutoDock, GraphPad Prism, RStudio (beginner), scRNA-seq, AI-assisted scripting, and workflow automation (*Hands-on beginner with strong conceptual understanding; comfortable learning and executing bioinformatics workflows under guidance.*)

➢ **Languages**
English (fluent), Hindi (native), Marathi & Telugu (limited working proficiency).

Presentations, Leadership, and Achievements

- Poster presentation: Asia Chromatin Meeting 2025, IISER Pune, Maharashtra, India.
- Oral Presentation: National Science Convention 2023, Osmania University, Hyderabad, India.
- Served as Biotechnology Branch Coordinator in the Training and Placement Cell at NIT-B.
- Qualified **GATE 2023** (Biotechnology), ranked within the top 10% nationwide. (Rank-1197)

References

Dr. Rupinder Kaur
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Dr. Radhika Tippani
Department of Biotechnology, Kakatiya University, Warangal, India (PIN 506002) | Email: tippanira@gmail.com